Project 02
AI Music & Audio Analyzer

Living Project Documentation — v0.4

Item	Current Value
Project Status	In Progress — Audio Capture, Signal & Spectral Analysis Verified
Platform	Windows Laptop
Python	3.14.7
Environment	VS Code + PowerShell + Python venv
Hardware	Windows Laptop + Airdopes Joy Bluetooth Earbuds
Audio Input	Airdopes Joy Hands-Free microphone
Audio Output	Airdopes Joy Stereo
Analysis Libraries	sounddevice + Librosa + NumPy + Matplotlib
Test Recording	5-second mono WAV — airdopes_test.wav

1. Project Definition

AI Music & Audio Analyzer is a Python-based project for real hardware audio capture, digital signal processing, visualization, frequency analysis, spectral feature extraction, and later AI interpretation. The Windows laptop is the processing platform and Airdopes Joy Bluetooth earbuds provide the microphone input and playback path.

2. Current Architecture

Airdopes Joy Microphone → Bluetooth → Windows Audio → Python sounddevice → WAV Capture → Librosa / NumPy → Time/Frequency/Spectral Analysis → Matplotlib Reports → Future Music Analysis / AI / Dashboard

3. Completed Milestones

Step	Milestone	Status
1	Audio device detection	PASS
2	Airdopes microphone verification	PASS
3	Five-second microphone recording	PASS
4	WAV loading and basic analysis	PASS
5	Waveform visualization	PASS
6	Audio level analysis	PASS
7	FFT / frequency spectrum analysis	PASS
8	STFT / spectrogram analysis	PASS
9	Spectral feature analysis	PASS

4. Step 1 — Audio Device Detection

Windows audio-device enumeration successfully detected the Airdopes Joy Bluetooth endpoints. Device 2 was identified as the Airdopes Joy Hands-Free microphone with one input channel and a reported 44.1 kHz rate. Device 4 was identified as Airdopes Joy Stereo with two output channels at 44.1 kHz.

5. Step 2 — Airdopes Microphone Verification

The Airdopes Joy Hands-Free microphone was selected through Python sounddevice and produced a non-zero signal during the microphone test.

6. Step 3 — Airdopes Audio Recording

A five-second mono recording was successfully captured from device 2 and saved as audio\airdopes_test.wav.

Parameter	Result
Device ID	2
Input Channels	1
Sample Rate	44,100 Hz
Duration	5.00 seconds
Samples	220,500
RMS Amplitude	0.031082
Peak Amplitude	0.423889

7. Step 4 — WAV Audio Analysis

The saved WAV file was loaded successfully with Librosa. The measured values matched the recording stage, confirming reliable file processing.

Sample Rate: 44,100 Hz **|** Channels: 1 **|** Samples: 220,500 **|** Duration: 5.00 sec

RMS: 0.031082 **|** Peak: 0.423889 **|** RMS: -30.15 dBFS **|** Peak: -7.45 dBFS

8. Step 5 — Waveform Visualization

The recorded signal was converted into a time-domain waveform with Librosa and Matplotlib. The graph was saved as reports\airdopes_waveform.png.

9. Step 6 — Audio Level Analysis

The audio-level analyzer calculated amplitude statistics, RMS, peak, dBFS levels, crest factor, and RMS level over time.

Measurement	Result
Minimum Amplitude	-0.347687
Maximum Amplitude	0.423889
Mean Absolute Amplitude	0.010442
RMS Amplitude	0.031082
Peak Amplitude	0.423889
RMS Level	-30.15 dBFS
Peak Level	-7.45 dBFS
Crest Factor	13.638
Level Classification	Healthy
Graph	reports\airdopes_audio_level.png

10. Step 7 — FFT / Frequency Spectrum Analysis

A Fast Fourier Transform was applied to the five-second Airdopes recording after applying a Hann window. The frequency spectrum was calculated and visualized.

FFT Measurement	Actual Result
Sample Rate	44,100 Hz
Samples	220,500
FFT Output Samples	110,251
Frequency Resolution	0.2000 Hz
Maximum Frequency / Nyquist	22,050 Hz
Dominant Frequency	544.60 Hz
Dominant Magnitude	0.001539
Graph	reports\airdopes_frequency_spectrum.png

The strongest detected FFT component was approximately 544.60 Hz. This is a characteristic of the particular microphone-test recording and is not by itself a music-pitch classification.

11. Step 8 — STFT / Spectrogram Analysis

Short-Time Fourier Transform was applied to observe how frequency content changes over time.

STFT Measurement	Actual Result
Sample Rate	44,100 Hz
FFT Size	2048
Hop Length	512
Frequency Bins	1025
Time Frames	431
Frequency Resolution	21.53 Hz
Maximum Frequency	22,050 Hz
Graph	reports\airdopes_spectrogram.png

The displayed spectrogram is limited to 0–10 kHz for easier inspection while the underlying STFT covers the 22.05 kHz Nyquist range.

12. Step 9 — Spectral Feature Analysis

Spectral features were extracted from the same Airdopes recording using Librosa. These features begin the transition from general signal processing toward music/audio characterization.

Feature	Actual Result
Spectral Centroid	7,159.79 Hz
Spectral Bandwidth	4,349.76 Hz
Spectral Rolloff (85%)	12,150.42 Hz
Spectral Flatness	0.329063
Zero Crossing Rate	0.147712
Brightness Classification	Bright
Texture Classification	Noise-like
Feature Graph	reports\airdopes_spectral_features.png

Interpretation note: the current recording is a short microphone test rather than a controlled music sample. Therefore the Bright / Noise-like classification is treated as an observation of this test signal, not as a final classification of music.

13. Current Analysis Outputs

Output	Purpose	Status
audio\airdopes_test.wav	Raw captured microphone recording	Generated
reports\airdopes_waveform.png	Amplitude versus time	Generated
reports\airdopes_audio_level.png	RMS level versus time	Generated
reports\airdopes_frequency_spectrum.png	Frequency versus magnitude	Generated
reports\airdopes_spectrogram.png	Frequency versus time	Generated
reports\airdopes_spectral_features.png	Spectral feature visualization	Generated

14. Current Project Structure

Project-02-AI-Music-Audio-Analyzer/

■■■ audio/ → airdopes_test.wav

■■■ reports/ → waveform, audio-level, frequency-spectrum, spectrogram and spectral-feature images

■■■ src/ → audio_devices.py, test_airdopes_mic.py, record_airdopes.py, analyze_airdopes_audio.py, plot_airdopes_waveform.py, audio_level_analysis.py, fft_frequency_analysis.py, spectrogram_analysis.py, spectral_features.py

■■■ tests/

■■■ venv/

■■■ requirements.txt

■■■ README.md

15. Testing Record

Test	Result
Audio device enumeration	PASS
Airdopes Joy microphone detection	PASS
Airdopes Joy stereo output detection	PASS
Microphone signal capture	PASS
Five-second WAV generation	PASS
WAV loading with Librosa	PASS
Waveform generation	PASS
Audio level analysis	PASS
RMS level graph generation	PASS
FFT / frequency spectrum	PASS
Dominant frequency extraction	PASS — 544.60 Hz
STFT / spectrogram	PASS
Spectral feature extraction	PASS

16. Next Development Step

Step 10 — Bass / Mid / Treble Energy Analysis. The next module will divide the frequency spectrum into practical audio bands and calculate their energy levels. This will move the project closer to music-oriented analysis.

17. Version History

v0.1 — 19-Aug-2026: New documentation baseline created from the beginning. Hardware setup, Python environment, device detection, microphone verification and recording documented.

v0.2 — 19-Aug-2026: Steps 4–6 completed. WAV loading, waveform visualization, RMS/peak/dBFS analysis, amplitude statistics, crest factor and RMS level graph documented.

v0.3 — 19-Aug-2026: Steps 7–8 completed. FFT frequency spectrum and STFT spectrogram documented with actual measurements and generated outputs.

v0.4 — 19-Aug-2026: Step 9 completed. Spectral centroid, bandwidth, rolloff, flatness, zero-crossing rate and basic sound-characteristic classifications documented with actual measured results.

Only the new Project-02 history is used; previous report versions are intentionally excluded.